

Quadrilaterals: Which Way Does the Implication Go?

Answer Key

High School Geometry — Polygons

Quick Answers

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|---------------|---------------|---------------------|
| 1. 15 | 3. — | 5. 8 |
| 2. 2.83, 8.49 | 4. 3.61, 6.71 | 6. 4.47, 6.32, 6.32 |
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Common wrong answers

2. If they got 8.49: assumed a rhombus has congruent diagonals, so reported AC 's length for BD
4. If they got 6.71: assumed perpendicular diagonals force all four sides congruent, so reported AB 's length for BC
5. If they got 3: subtracted the 5 instead of adding it when moving it across the equals sign
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Part (a) of each problem is a computation and is machine-verified. Part (b) is a diagnosis in words: a model answer is given, together with what to accept and what not to. The sentence to insist on names *both* the property that was assumed and the property that was actually established.

1. $MNPQ$ is a rhombus with consecutive angles $(4x + 10)^\circ$ and $(6x + 20)^\circ$.

(a) **Solution:** A rhombus is a parallelogram, so consecutive angles are supplementary.

$$(4x + 10) + (6x + 20) = 180$$

$$10x + 30 = 180$$

$$x = 15$$

$x = 15$

The angles are $4(15) + 10 = 70^\circ$ and $6(15) + 20 = 110^\circ$.

(b) **Manual review — expected reasoning.** No. A square is a rhombus with right angles, so the implication square $\Rightarrow$ rhombus runs one way only. This rhombus has a 70° angle, so it is not a square. *Accept* any answer that points at the angles (or at the diagonals not being congruent). *Do not accept* “no, because it does not look like one” or an answer that treats “all sides congruent” as decisive.

2. $A(1, 1)$, $B(5, 3)$, $C(7, 7)$, $D(3, 5)$, all four sides congruent. Priya: “a square, so both diagonals are about 8.49”.

(a) **Solution:** Use the distance formula on each diagonal.

$$AC = \sqrt{(7-1)^2 + (7-1)^2} = \sqrt{72} \quad BD = \sqrt{(3-5)^2 + (5-3)^2} = \sqrt{8}$$

$AC \approx 8.49$

$BD \approx 2.83$

(b) **Manual review — expected reasoning.** Priya's first step is sound: four congruent sides do force a rhombus. The step that fails is the next one — rhombus $\Rightarrow$ square, which is the converse of a one-way implication. The diagonals settle it: $8.49 \neq 2.83$, so $ABCD$ is not a rectangle and therefore not a square. Most specific correct name: **rhombus**. *Do not accept* "Priya is right, she just miscalculated" — the arithmetic was never the problem.

3. Always true or not always true, with a counterexample where it fails.

Manual review — expected answers.

(a) **Always true.** A rectangle has two pairs of parallel sides by definition, which is the definition of a parallelogram. (This is the "upward" direction of the family tree.)

(b) **Not always true.** A parallelogram needs right angles to be a rectangle. Counterexample: the parallelogram with vertices $(0, 0)$, $(5, 0)$, $(7, 3)$, $(2, 3)$ — opposite sides are parallel and congruent, but no angle is 90° .

(c) **Not always true.** Perpendicular diagonals give a rhombus only if the diagonals also *bisect* each other. Counterexample: the kite with vertices $(0, 5)$, $(2, 0)$, $(0, -3)$, $(-2, 0)$ — its diagonals lie on the axes and so are perpendicular, but the sides measure $\sqrt{29}$ and $\sqrt{13}$, so they are not all congruent.

(d) **Always true.** A square is by definition a quadrilateral with four congruent sides and four right angles; four congruent sides alone is the definition of a rhombus.

Accept any correct counterexample (a described drawing is fine). *Do not accept* a bare "false" with no counterexample for (b) or (c).

4. $A(0, 6)$, $B(3, 0)$, $C(0, -2)$, $D(-3, 0)$. Sam: "perpendicular diagonals, so a rhombus with every side about 6.71".

(a) **Solution:**

$$AB = \sqrt{(3-0)^2 + (0-6)^2} = \sqrt{45} \quad BC = \sqrt{(0-3)^2 + (-2-0)^2} = \sqrt{13}$$

$$AB \approx 6.71$$

$$BC \approx 3.61$$

(b) **Manual review — expected reasoning.** Sam is right that the diagonals are perpendicular, but that alone is not sufficient: for a rhombus the diagonals must be perpendicular *and* bisect each other. Here the diagonal AC has midpoint $(0, 2)$ while BD has midpoint $(0, 0)$, so they do not bisect each other — and indeed $AB \neq BC$. $ABCD$ is a **kite**. *Accept* either the midpoint argument or the two unequal side lengths as the disproof; the sentence must say what perpendicular diagonals alone do *not* guarantee.

5. $PQRS$ is a parallelogram, $PR = 3x - 5$, $QS = x + 11$. Sam gets $x = 3$ and says "square".

(a) **Solution:** Setting the diagonals equal is the right move; the slip is in the algebra. Moving -5 across the equals sign *adds* 5:

$$3x - 5 = x + 11 \Rightarrow 2x = 11 + 5 = 16 \Rightarrow x = \frac{16}{2}$$

$$x = 8$$

Sam computed $11 - 5 = 6$ and halved it, which is where the 3 came from. Check: both diagonals measure $3(8) - 5 = 19$ and $8 + 11 = 19$.

(b) **Manual review — expected reasoning.** Congruent diagonals in a *parallelogram* force a **rectangle** — that theorem is genuine. But rectangle $\Rightarrow$ square is the converse of

a one-way implication and does not follow: a 19-unit diagonal says nothing about the side lengths. To reach “square” you would need one more fact, such as two congruent adjacent sides or perpendicular diagonals. *Do not accept* “Sam is right once x is fixed”.

6. Challenge: $W(-1, 2)$, $X(3, 4)$, $Y(5, 0)$, $Z(1, -2)$, all four sides congruent.

(a) Solution:

$$WX = \sqrt{4^2 + 2^2} = \sqrt{20} \quad WY = \sqrt{6^2 + (-2)^2} = \sqrt{40} \quad XZ = \sqrt{(-2)^2 + (-6)^2} = \sqrt{40}$$

$WX \approx 4.47$

$WY \approx 6.32$

$XZ \approx 6.32$

(b) Manual review — expected reasoning. Four congruent sides give a rhombus; congruent diagonals in a parallelogram give a rectangle; a quadrilateral that is both is a **square**. Unlike problem 2, the diagonal test passes here, and that single test — comparing WY with XZ — is exactly the one that separates “rhombus but not square” from “square”. An equivalent sufficient test is showing one angle is right, e.g. the slopes of WX and XY are $\frac{1}{2}$ and -2 . *Accept* either test; the answer must name the test, not just the shape.